USER INTERFACE DESIGN EX-3

Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)

NAME:SANGAMITHRA ARUN

RNO:240701465

Command line Interface(CLI)

● A Text-based Interface (Command line interface) enables users to communicate with a computer using the command line, as opposed to using a graphical interface.

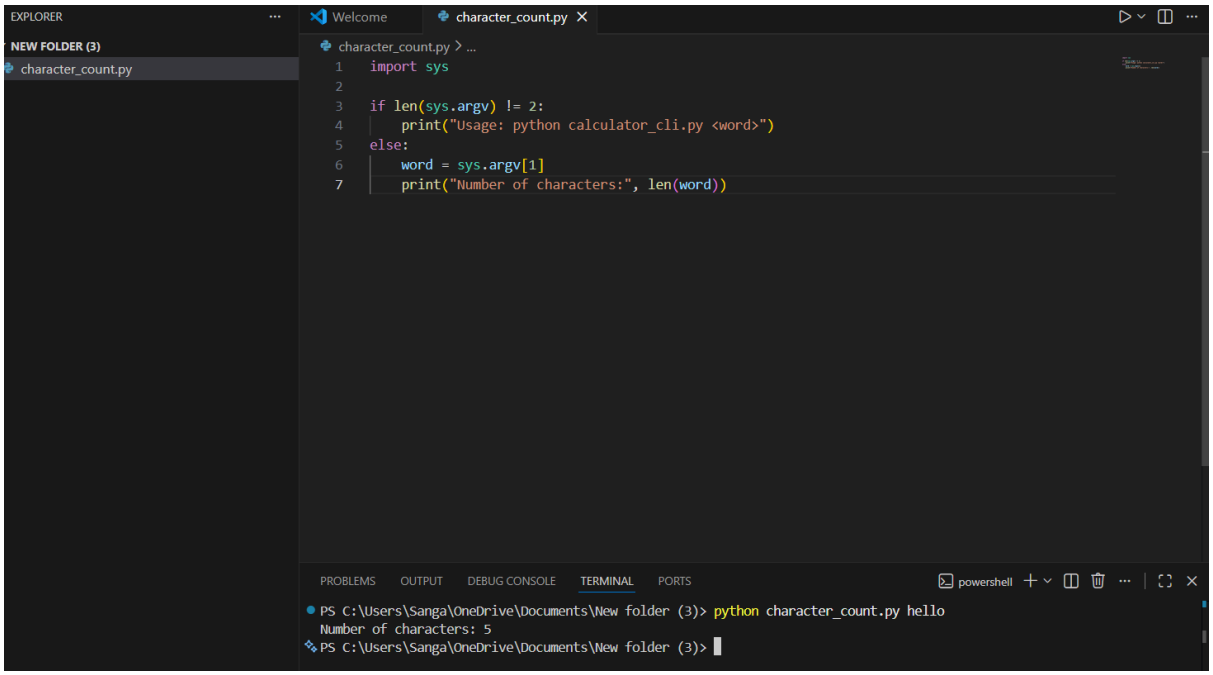
● Compact and Powerful – CLI is usually faster and more powerful than a GUI, particularly for expert users who know the correct commands.

● Automation & Scripting – The command-line interface is there to allow users to automate tasks with scripts, making the repetitive tasks easier.

● Lightweight and Resource Friendly – Compared to GUI application, CLI uses very little system resources which is why they are most suited for use on remote servers

example: Windows command prompt

CLI :IMPLEMENTATION AND OUTPUT



GUI:

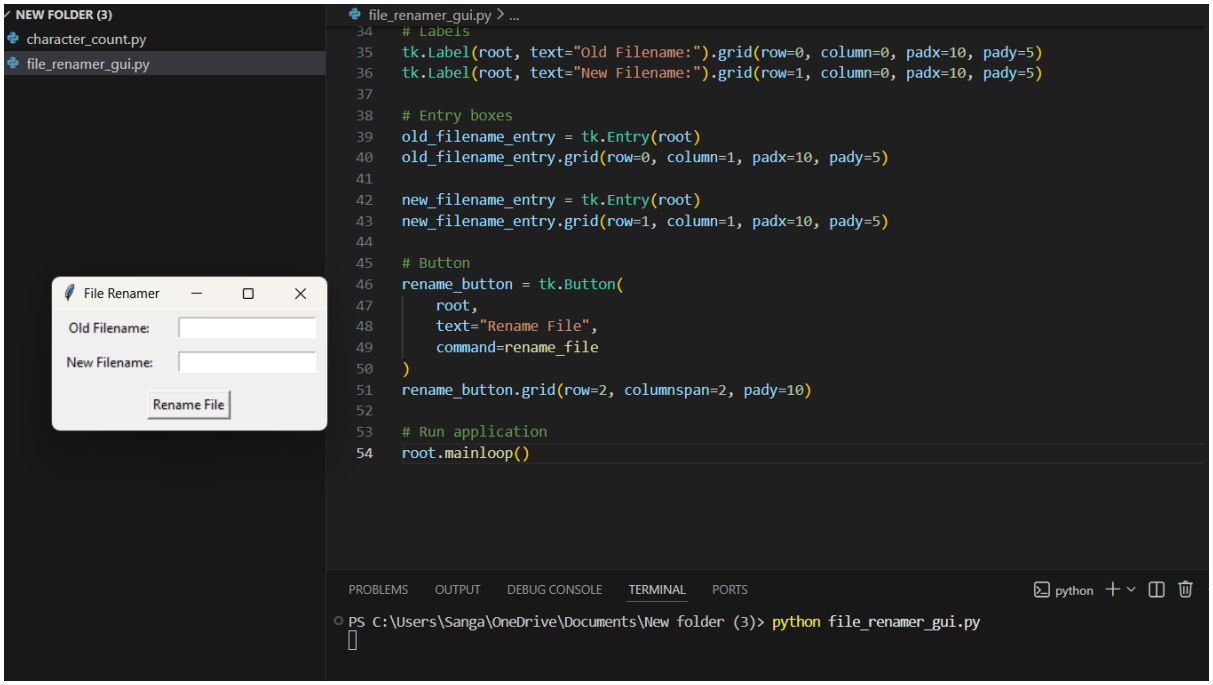
● A GUI uses visual elements including icons, buttons, and windows which makes it more user-friendly for users to work with the system.

● More Resource Intensive – GUI applications use more system resources (CPU, RAM, and GPU) in contrast to a Command Line Interface (CLI) that can degrade performance in low-end devices.

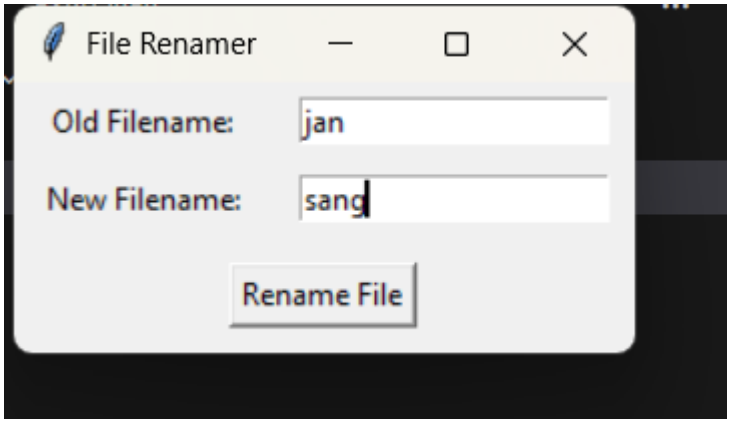
● Limited Automation: GUI does not have automation and scripting abilities like the CLI and is thus more tedious for repetitive jobs.

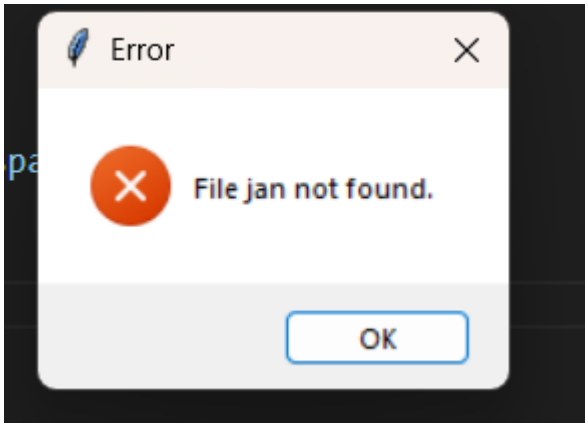
example: Windows, Android, iOS

IMPLEMENTATION:



OUTPUT:





Voice user interface(VUI)

• No Hands – VUI allows me to talk to a system and in some cases this is much more convenient than having to type. • Natural Communication– VUI allows for more natural and fluid communication, as it can process users' speech and verbal commands, eliminating typing or clicking.

• Accuracy Challenges – Speech recognition can occasionally have difficulty with accents, background noise, and complex commands, sometimes resulting in misinterpretation.

example: Amazon, Alexa, Google Assistant

IMPLEMENTATION:

```python
def listen(prompt):
        text = recognizer.recognize_google(audio)
        print("You said:", text)
        return text
    except:
        print("Could not understand audio")
        return None


# Main program
print("🎤 Welcome to Voice Controlled File Renamer")

old_file = None
while old_file is None:
    old_file = listen("Say the OLD file name (without txt)")

new_file = None
while new_file is None:
    new_file = listen("Say the NEW file name (without txt)")

rename_file(old_file, new_file)
```

OUTPUT:

```
PS C:\Users\Sanga\OneDrive\Documents\uid vui> python voice_file_rename.py
🎤 Welcome to Voice Controlled File Renamer
Say the OLD file name (without txt)
Could not understand audio
Say the OLD file name (without txt)
You said: old
Say the NEW file name (without txt)
Could not understand audio
Say the NEW file name (without txt)
Could not understand audio
Say the NEW file name (without txt)
You said: new new
✅ File renamed from old.txt to new new.txt
```